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PROJECT ON

Restaurant & Consumer Data Analysis

SQL Analysis & Business Insights

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About Me

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- **Education:** B.Tech Computer Science Engineering (Artificial Intelligence & Machine Learning)
- **Career Goal:** Aspiring Data Analyst passionate about SQL, Python, Power BI and Data Visualization
- **Experience:** Fresher | Academic SQL Projects | GitHub Portfolio
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Agenda

- Project Objective
- Database Design & SQL Schema
- SQL Concepts Used
- Business Analysis Questions
- Advanced SQL Implementation
- Key Insights & Conclusion
- Challenges, Learning Outcomes & Q&A

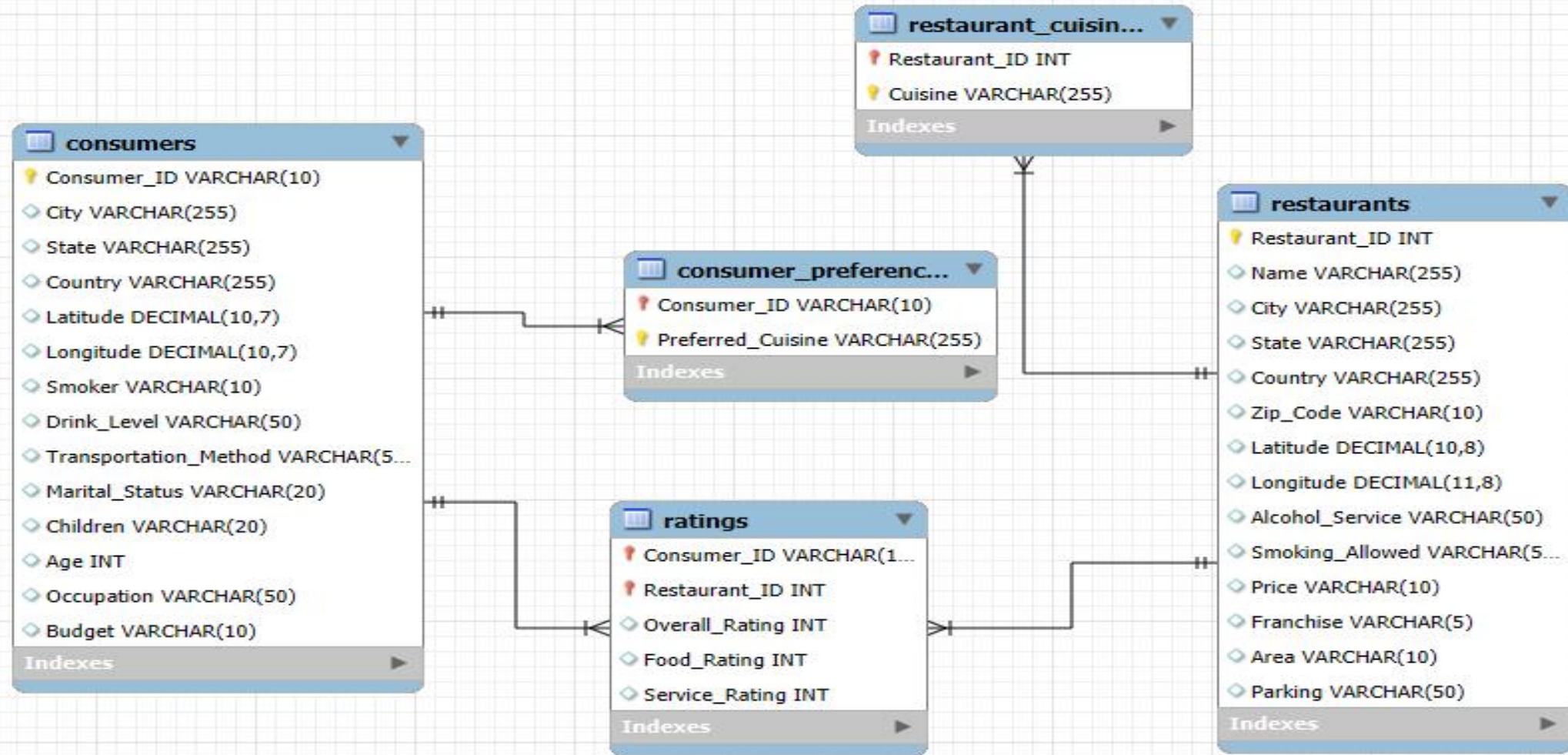
Project Objective

- **Goal:** Build a structured SQL database from multiple CSV files covering consumers, restaurants, cuisines and ratings
- Database Design & Table Relationships
- Data Cleaning
- SQL Query Writing (Basic to Advanced)
- Business Analytics & Reporting
- Customer Behaviour Analysis
- Restaurant Performance Analysis

Dataset Overview

- **Consumers:** demographic & lifestyle details of each customer
- **Consumer Preferences:** preferred cuisine for each consumer
- **Restaurants:** restaurant details - city, price, franchise, parking, alcohol service
- **Restaurant Cuisines:** cuisine type served by each restaurant
- **Ratings:** overall, food & service ratings given by consumers
- **Total Tables: 5** | Database Engine: MySQL

ER Diagram



Database Design

- **Relational Flow:** Consumers -> Ratings -> Restaurants -> Restaurant Cuisines & Consumer Preferences
- **Primary Keys:** Consumer_ID, Restaurant_ID uniquely identify each record
- **Foreign Keys:** link Ratings to Consumers and Restaurants for referential integrity
- **Normalization:** tables structured to avoid redundancy and maintain consistency
- Relational design enables multi-table joins for deep business analysis

SQL Skills Used

- **DDL:** CREATE DATABASE, CREATE TABLE, ALTER TABLE, PRIMARY KEY, FOREIGN KEY
- **DML:** SELECT, WHERE, GROUP BY, HAVING, ORDER BY, INSERT
- **Joins:** INNER JOIN, LEFT JOIN, RIGHT JOIN across multiple tables
- **Subqueries & Derived Tables:** for layered filtering and analysis
- **Advanced SQL:** CTEs, Views, Stored Procedures, Window Functions
- **Functions:** CASE, ROW_NUMBER(), RANK(), DENSE_RANK(), AVG(), COUNT(), SUM(), MAX(), MIN()

Business Questions Solved

- **Answered 35+ real-world SQL business questions, including:**
- Highest-rated and top-performing restaurants
- Customer cuisine preferences & budget segmentation
- Franchise vs. non-franchise restaurant analysis
- Cuisine popularity and rating trends across cities
- Consumer engagement and behaviour patterns
- Ranking restaurants by rating using window functions

Advanced SQL Features

- **CTEs:** structured multi-step filtering, e.g. identifying consumers by city before joining ratings
- **Window Functions:** RANK(), DENSE_RANK(), ROW_NUMBER(), LEAD() for ranking and trend analysis
- **Views:** HighlyRatedMexicanRestaurants, ConsumerAverageRatings for reusable logic
- **Stored Procedures:** GetRestaurantRatingsAboveThreshold, GetConsumerSegmentAndRestaurantPerformance
- **Derived Tables & CASE Logic:** used for segmentation and performance flagging

Sample SQL Queries

- Sample Data:

```
502
503 • select * from consumers limit 5;
```

Consumer_ID	City	State	Country	Latitude	Longitude	Smoker	Drink_Level	Transportation_Method	Marital_Status	Children	Age	Occupation	Budget
U1001	San Luis Potosi	San Luis Potosi	Mexico	22.139997	-100.978803	No	Abstemious	On Foot	Single	Independent	23	Student	Medium
U1002	San Luis Potosi	San Luis Potosi	Mexico	22.150087	-100.983325	No	Abstemious	Public	Single	Independent	22	Student	Low
U1003	San Luis Potosi	San Luis Potosi	Mexico	22.119847	-100.946527	No	Social Drinker	Public	Single	Independent	23	Student	Low
U1004	Cuernavaca	Morelos	Mexico	18.867	-99.183	No	Abstemious	Public	Single	Independent	72	Employed	Medium
U1005	San Luis Potosi	San Luis Potosi	Mexico	22.183477	-100.959891	No	Abstemious	Public	Single	Independent	20	Student	Medium

consumers 2 x

```
504
505 • select * from restaurants limit 5;
```

Restaurant_ID	Name	City	State	Country	Zip_Code	Latitude	Longitude	Alcohol_Service	Smoking_Allowed	Price	Franchise	Area	Parking
132560	Puesto de Gorditas	Ciudad Victoria	Tamaulipas	Mexico		23.7523041	-99.1669133	None	Yes	Low	No	Open	Public
132561	Cafe Ambar	Ciudad Victoria	Tamaulipas	Mexico		23.726819	-99.1265059	None	No	Low	No	Closed	None
132564	Church's	Ciudad Victoria	Tamaulipas	Mexico		23.7309245	-99.1451848	None	No	Low	No	Closed	None
132572	Cafe Chaires	San Luis Potosi	San Luis Potosi	Mexico		22.1416471	-100.9927118	None	No	Low	No	Closed	Yes
132583	McDonalds Centro	Cuernavaca	Morelos	Mexico	62000	18.9222904	-99.234332	None	No	Low	Yes	Closed	None

```
507 • select * from ratings limit 5;
```

Consumer_ID	Restaurant_ID	Overall_Rating	Food_Rating	Service_Rating
U1077	135085	2	2	2
U1077	135038	2	2	1
U1077	132825	2	2	2
U1077	135060	1	2	2
U1068	135104	1	1	2

```
508
509 • select * from consumer_preferences limit 5;
```

Consumer_ID	Preferred_Cuisine
U1001	American
U1002	Mexican
U1003	Mexican
U1004	Bakery
U1004	Breakfast

Questions JOINS with Subqueries

```
104  /* 6.Find consumers (Consumer_ID, Age, Occupation) who have rated at least one restaurant  
105  but have NOT rated any restaurant that serves 'Italian' cuisine.*/  
106  •  SELECT DISTINCT c.Consumer_ID, c.Age, c.Occupation  
107  FROM consumers c  
108  JOIN ratings r  
109  ON c.Consumer_ID = r.Consumer_ID  
110  WHERE c.Consumer_ID NOT IN (  
111  SELECT r2.Consumer_ID  
112  FROM ratings r2  
113  JOIN restaurant_cuisines rc  
114  ON r2.Restaurant_ID = rc.Restaurant_ID  
115  WHERE rc.Cuisine = 'Italian'  
116  );  
117
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

Consumer_ID	Age	Occupation
U1077	25	Student
U1068	24	Student
U1067	25	Student
U1103	23	Employed
U1107	23	
U1044	43	Employed

Advanced SQL Concepts: Derived Tables

```
238
239  /* 2.For each Occupation, find the average age of consumers.
240     Only consider consumers who have made at least one rating.
241     (Use a derived table to get consumers who have rated).*/
242  •  SELECT Occupation, AVG(Age) AS avg_age
243     FROM (
244         SELECT DISTINCT c.Consumer_ID, c.Age, c.Occupation
245         FROM consumers c
246         JOIN ratings r
247             ON c.Consumer_ID = r.Consumer_ID
248     ) AS rated_consumers
249     GROUP BY Occupation;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
Occupation	avg_age			
Student	25.1327			
Employed	37.5625			
	40.4286			
Unemployed	21.5000			

Advanced SQL Concepts: CTE

```
221  /* 1.Using a CTE, find all consumers who live in 'San Luis Potosi'.
222  Then, list their Consumer_ID, Age, and the Name of any Mexican restaurant they have rated with an Overall_Rating of 2.*/
223  WITH slp_consumers AS (
224      SELECT Consumer_ID, Age
225      FROM consumers
226      WHERE City = 'San Luis Potosi'
227  )
228  SELECT DISTINCT c.Consumer_ID, c.Age, r.Name
229  FROM slp_consumers c
230  JOIN ratings rt
231      ON c.Consumer_ID = rt.Consumer_ID
232  JOIN restaurants r
233      ON rt.Restaurant_ID = r.Restaurant_ID
234  JOIN restaurant_cuisines rc
235      ON r.Restaurant_ID = rc.Restaurant_ID
236  WHERE rc.Cuisine = 'Mexican'
237  AND rt.Overall_Rating = 2;
```

Result Grid			
Filter Rows:			
Export: 			
Wrap Cell Content: ☐			
	Consumer_ID	Age	Name
▶	U1137	23	Gordas De Morales
	U1108	29	Gordas De Morales
	U1095	26	Gordas De Morales
	U1061	22	Gordas De Morales
	U1048	24	Gordas De Morales
	U1003	23	Gordas De Morales
	U1134	21	Cabana Huasteca

Advanced SQL Concepts: VIEW

```
307  /* 7.Create a VIEW named HighlyRatedMexicanRestaurants that shows the Restaurant_ID,  
308  Name, and City of all Mexican restaurants that have an average Overall_Rating greater than 1.5.*/  
309  • create view highlyratedmexicanrestaurants as  
310  select  
311      r.restaurant_id,  
312      r.name,  
313      r.city  
314  from restaurants r  
315  join ratings rt  
316      on r.restaurant_id = rt.restaurant_id  
317  join restaurant_cuisines rc  
318      on r.restaurant_id = rc.restaurant_id  
319  where rc.cuisine = 'Mexican'  
320  group by r.restaurant_id, r.name, r.city  
321  having avg(rt.overall_rating) > 1.5;  
322  • select * from highlyratedmexicanrestaurants;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

restaurant_id	name	city
135028	La Virreina	San Luis Potosi
134976	Log Yin	Cuernavaca
135025	El Rincon De San Francisco	San Luis Potosi
132755	La Estrella De Dimas	San Luis Potosi
135055	La Cochinita Pibil Restaurante Yucateco	San Luis Potosi

Advanced SQL Concepts: Stored Procedure

```
341  /* 9.Create a stored procedure GetRestaurantRatingsAboveThreshold that accepts a Restaurant_ID and a minimum Overall_Rating as input. It should return the Consumer_ID,  
342  Overall_Rating, Food_Rating, and Service_Rating for that restaurant  
343  where the Overall_Rating meets or exceeds the threshold.*/  
344  delimiter //  
345  
346  • create procedure GetRestaurantRatingsAboveThreshold(  
347      in p_restaurant_id int,  
348      in p_min_rating int  
349  )  
350  • begin  
351      select  
352          consumer_id,  
353          overall_rating,  
354          food_rating,  
355          service_rating  
356      from ratings  
357      where restaurant_id = p_restaurant_id  
358          and overall_rating >= p_min_rating;  
359  end //  
360  delimiter ;  
361  • CALL GetRestaurantRatingsAboveThreshold(132561, 1);  
362
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	consumer_id	overall_rating	food_rating	service_rating
▶	U1026	1	1	1
	U1129	1	2	2
	U1065	1	1	1

Key Findings

- Restaurant ratings vary significantly across cuisine types
- Consumer preferences strongly influence overall ratings given
- Budget category (Low/Medium/High) impacts restaurant choice
- Mexican cuisine appears frequently in consumer preferences
- Higher-rated restaurants perform consistently well on both food and service ratings
- SQL can generate actionable business insights from structured relational data

Challenges Faced

- Designing a clean relational schema from raw CSV files
- Understanding and enforcing foreign key relationships
- Writing complex multi-table joins
- Building and debugging CTEs and nested subqueries
- Debugging stored procedures with conditional logic
- Optimizing SQL queries for performance

Learning Outcomes

- SQL Query Writing (Basic to Advanced)
- Window Functions (RANK, ROW_NUMBER, LEAD)
- CTEs & Derived Tables
- Views & Stored Procedures
- Business Analytics & Data Interpretation

Conclusion

- Transformed raw restaurant and consumer datasets into a structured relational database
- Generated meaningful business insights using filtering, joins, aggregations, CTEs, views, stored procedures and window functions
- Demonstrates strong SQL problem-solving skills applicable to real-world Data Analyst roles

My Experience

- Working on this project significantly improved my SQL knowledge
- I learned how to design databases, write optimized queries, and solve real-world business problems
- Implemented advanced SQL concepts: CTEs, Views, Stored Procedures and Window Functions
- This project increased my confidence in handling real business datasets

THANK YOU

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